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MILITARY SPECIFICATION

MANIPULATORS, AUTOMATIC WELDING HEAD, STATIONARY BASE, AND TRAVELING RAM OR BOOM, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven manipulators with a stationary base and a traveling boom for accomplishing versatile and complex manipulation for use with automatic welding heads. Welding heads are not covered by this specification.

1.2 Classification. The manipulator shall be of the sizes listed in table I. The size to be furnished shall be as specified (see 3.5 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, TN 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this Document or by letter.

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SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

OSHA 29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete manipulator shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The manipulator shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The manipulator shall be capable of transporting an automatic welding head and associated equipment along a required path for performing horizontal and circumferential welding operations in accordance with the requirements herein. The manipulator shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein, but are a part of the manufacturer's current design, shall be included in the manipulator furnished. The manipulator shall be designed to incorporate a stationary base and column with 360 degrees manual or power operated rotation of the column, a ram or boom, with manual or powered lift and manual or powered reach travel as specified herein. The manipulator shall be designed with controls for selection of rate of horizontal motion of the welding head with reversible variable speed silicon control rectifier (SCR) direct current (DC) motor drive with adjustment of rate within the capacity of the manipluator. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

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3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the manipulator shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate manipulator movement such as lift and reach travel shall be graduated in increments of inches per minute (IPM). Table rotation shall be graduated in increments not larger than 1 degree. Dial, handwheel or handcrank circumferences and dial or controls position indexing shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.2 Reclaimed materials. The manipulator may contain reclaimed materials to the maximum extent possible providing such materials will not jeopardize the intended use, performance, or design life of the manipulator. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The manipulator and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, electronic and electrical operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the manipulator, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Noise level. Unless otherwise specified (see 6.2), noise level emitted by the manipulator when operating under a no load condition shall not exceed 84 decibel (db) when measured on the A scale of a type I or type II sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the manipulator, and its components, the manipulator shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the manipulator at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

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3.2.7 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or manipulator.

3.3 Construction. The manipulators shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the manipulator at its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the manipulator and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have a hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The manipulator shall consist of not less than the stationary base, column, crosshead, ram or boom, welding accessories, and all necessary controls.

3.4.1 Stationary base. The stationary base shall function as support for all components of the manipulator and shall provide 360 degree rotation of the column. The base shall be constructed of either fabricated steel or steel structural shape. Unless otherwise specified (see 6.2), the base shall be of the free standing type and shall have radially extended legs for rigidity. Portability and leveling shall be accomplished by casters, locking devices, and jack screws. When specified (see 6.2), the base shall be of the bolt-down type with bolt holes for mounting the base to the floor. Unless otherwise specified (see 6.2), the base shall support the column with a bearing type arrangement allowing smooth column rotation.

3.4.2 Column. The column shall provide rotary and vertical motion of the crosshead. The column shall also support the control center with connections to the electrical power supply and welding accessories. Rotary motion of the column shall be accomplished with either a manual handcrank or SCR constant or variable speed drive system. When only one drive system for rotary motion of the column is acceptable, the particular drive required shall be as specified (see 6.2). When specified (see 6.2), the rotating column shall be provided with a reversible, infinitely variable speed silicon-control rectifier (SCR) direct current (DC) motor drive. The rate of infinitely variable rotational speed SCR drive shall be not less than 0.007 revolutions per minute (RPM) and not more than 0.200 RPM. The rate of constant rotational speed shall be not more than 0.200 RPM. Provisions shall be made to lock the column at any rotary position. The vertical drive shall be mounted on the column or crosshead, and shall provide the lift as specified in table I.

3.4.3 Crosshead. The crosshead shall travel vertically on the column, and shall support horizontal motion of the ram or boom. Vertical motion of the crosshead on the column shall be on precision way surfaces suitable to provide the required performance. Vertical motion of the crosshead shall be accomplished with either a manual handcrank or SCR constant speed or infinitely variable speed drive system. When only one drive system for the vertical motion of the crosshead is acceptable, the particular drive required shall be as specified (see 6.2). When specified (see 6.2), the crosshead shall be provided with a reversible, infinitely variable speed silicon-controlled rectifier (SCR) direct current (DC) motor drive. The infinitely, variable speed shall be not more than 1.1 inches per minute (IPM) and not less than 34 IPM. When specified (see 6.2), the crosshead vertical motion shall be provided with chain drive capability. The chain drive system shall include a ball type anti-fall device. Lift gears or screw drives shall be provided with self-locking, anti-fall capabilities. The manipulator shall include multiple units, with each unit capable of supporting not less than the full rated load on the crosshead, or a single unit capable of supporting not less than twice the full rated load on the crosshead. The crosshead shall guide horizontal travel of the ram or boom through, across, or on the crosshead with reach conforming to the requirements of table I.

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Table I. Lift and reach.

Size	Ram or boom lift travel (feet)	Ram or boom reach travel (feet)	Size	Ram or boom lift travel (feet)	Ram or boom reach travel (feet)
1	4	4	19	12	15
2	5	5	20	12	16
3	6	6	21	14	11
4	6	7	22	14	14
5	6	8	23	14	15
6	4	10	24	15	12
7	8	6	25	15	15
8	8	8	26	15	20
9	8	10	27	16	16
10	8	12	28	16	20
11	9	7	29	18	18
12	9	9	30	20	20
13	9	14	31	20	24
14	10	10	32	24	24
15	10	12	33	24	30
16	10	14	34	25	25
17	12	12	35	30	30
18	12	14	36	35	35

3.4.3.1 Ram or boom. The ram or boom shall travel horizontally from the crosshead and shall support the total load of the automatic welding head and welding accessories, and components. The ram or boom lift and reach shall be as specified in table I.

3.4.3.2 Ram or boom lift . The lift of the ram or boom shall be either manually operated with worm and worm gear or constant or variable speed DC motor drive. When only one drive system is acceptable, on the ram or boom lift, the particular drive system required shall be as specified (see 6.2). Unless otherwise specified (see 6.2), the DC power operated variable speed drive shall provide infinite lift rates of travel from not more than 27 inches per minute (IPM) to not less than 100 IPM. Maximum allowable deviation of the lift rate of travel shall not exceed + 2 percent from the control set point for precision repeatability, and positionability, throughout the lift range specified in table I.

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3.4.3.3 Ram or boom reach. The boom reach shall be either manually operated with worm and worm gear or constant or variable speed DC motor drive. When only one drive system on the reach of the ram or boom is acceptable, the particular drive required shall be as specified (see 6.2). Unless otherwise specified (see 6.2), the DC power operated variable speed drive shall provide infinite reach rates of travel from not more than 6 inches per minute (IPM) to not less 60 IPM. The maximum allowable deviation of the reach rate of travel shall not exceed ± 2 percent of the control set point for precision repeatability, and positionability throughout the reach range specified in table I.

3.4.4 Accuracy. The accuracy rate of lift and reach motion of the manipulator shall be within ± two percent of the set rate throughout the full lift and reach traveling ranges with any load up to the maximum design load. Unless otherwise specified (see 6.2), the path of travel deviation shall be within ± 0.005 inch per linear foot. This requirement shall permit quality welding with the automatic submerged arc process on material thickness of 3/8 to 1-1/2 inches using mild steel and automatic precision metal inert gas (MIG) welding or tungsten inert gas (TIG) welding of materials ranging from 10 gauge to 1/2 inch in mild steel thicknesses.

3.4.5 Electrical requirements. The manipulator shall be wired to conform to NFPA 79. Each manipulator shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each manipulator shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-voltage. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The manipulator shall be initially wired as specified (see 6.2).

3.4.5.1 Motors. Motors shall meet the requirements of 3.4.5. Motors shall be rated for heavy duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure in accordance with the requirements of NEMA ICS 1 . All motors shall conform to the requirements of NEMA MG-1.

3.4.5.2 Controls. The controls shall conform to the requirements of NEMA ICS 1. The controls for operating the manipulator shall be supplied in a pendant type control with attached multi-conductor cable of not less than 20 feet in length. The controls for the manipulator shall consist of all devices required to operate the manipulator. The controls for the manipulator shall include an emergency stop button to stop all function of the manipulator. Unless otherwise specified (see 6.2), the controls shall include not less than the following:

a. Potentiometer type control for reach, lift, and rotational modes of motion.

b. Potentiometer type control for variable speeds.

c. Potentiometer type control for inching motion.

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d. Forward and reverse selector switch.

e. Emergency stop switch.

3.4.6 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the lift and reach of the manipulators boom shall be not less than those specified in table I, for the size machine specified. The load capacity for size 1 and 2 manipulators shall be not less than 300 pounds and for size 3 and larger units not less than 500 pounds.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the manipulator shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the manipulator will fit in its future operating location.

3.6 Performance. The manipulator shall be capable of traversing an automatic welding head and associated equipment with maximum load capacity along the reach and lift paths at the rates of motion specified in table I. Motion shall be smooth with no jerking or oscillating which will effect welding performance.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial manipulator shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment specified in the following subparagraphs shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or manipulator.

a. Indicating meters.

b. Operator's ladders, walkways, seats or workplate forms.

c. Welding head.

d. Welding head mount.

e. Welding equipment mounting.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.10 Marking on instruments, control panels, charts, and plates. All words on the manipulator including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each manipulator. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each manipulator. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Lift travel (feet)
Reach travel (feet)
Load capacity (pounds)
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Workmanship. Workmanship of the manipulator and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The manipulator shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each manipulator prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.10. The manipulator shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each manipulator to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the manipulator is installed at its final location. The acceptance test shall consist of the examination in 4.7, and all tests in 4.8. The manipulator shall pass the examination and all tests to be accepted.

4.7 Examination. Each manipulator shall be examined to determine compliance with all requirements of this specification.

4. 8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit (F) and 20 percent to 95 percent relative humidity.

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4.8.2 Operational test. Each manipulator shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance test. The manipulator shall be connected to the proper power source. The welding head shall be attached to the manipulator ram or boom, in its lowest and retracted position, and the manipulator shall be tested for conformance with 3.6. The welding head for the performance test shall be furnished by the contractor. All performance tests shall be completed without making any adjustments to or performing any maintenance on the manipulator.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.3.2 Rotary motion test. The manipulator, operating at its maximum rating and capacity load, shall be tested for rotary motion by operating the controls to raise the crosshead and loaded ram or boom three feet. The column shall then be rotated 180 degrees in each direction not less than three times. The rotary motion shall conform with the requirements of 3.4.1

4.8.3.3 Lift motion test. The manipulator, operating at its maximum rating and capacity load, shall be tested for lift motion by operating the controls to raise the crosshead and loaded ram or boom to the top of lift travel specified in table I. The load shall be lowered and raised not less than three times. The precision of variable rates and the minimum and maximum rates of lift motion shall conform with the requirements specified in 3.4.3.2.

4.8.3.4 Reach motion test. The manipulator, operating at its maximum rating and capacity load, shall be tested for reach by operating the controls to extend the loaded boom or ram to the end of the reach travel specified in table I. The boom or ram shall be retracted and extended not less than three times. Without distortion or malfunction. The rates and precision of variable rates and the maximum rate of reach motion shall conform to the requirements specified in 3.4.3.3.

4.8.3.5 Path-accuracy test. The manipulator, operating at its maximum rating and capacity load, shall be tested for accuracy of path travel by placing the crosshead at the bottom of lift and the ram or boom at retracted reach. With the welding head in position on the boom and the tip of the welding rod clamped at welding position, the controls shall be operated to extend the ram or boom to the full maximum reach. The manipulator shall be tested for conformance with the requirement 3.4.4.

4.8.3.6 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.1. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station of the manipulator. Noise emitted by the manipulator when operating under a no load condition shall not exceed 84 decibel db when measured on the A scale of a type I or type II sound level meter at slow response. The test shall be conducted with the manipulator operating at constant and variable speeds.

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4.8.3.8 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the manipulator offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.10 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The manipulators covered by this specification are intended for use in welding shops for the versatile and complex positioning of automatic welding heads in the performance of circumferential and longitudinal seam welds.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Noise level requirements, if different (see 3.2.5.1).

h. Environmental protection requirements exceeding EPA/OSHA requirements, if different (see 3.2.6).

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i. Specify surface finish requirements, if different (see 3.3.3).

j. Painting, if different (see 3.3.5.).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Base, free standing type, if different (see 3.4.1).

m. Base with bolt holes, if required (see 3.4.1).

n. Base support arrangement, if different (see 3.4.1).

o. When only one drive system for rotary motion of the column is acceptable, the particular drive required shall be as specified (see 3.4.2).

p. Variable rotational speed DC drive of the column, if required (see 3.4.2).)

q. When only one drive system for vertical motion of the crosshead on the column is acceptable, the particular drive required shall be as specified (see 3.4.3).

r. Variable speed DC drive of the crosshead, if required (see 3.4.3).

s. Crosshead vertical motion by chain drive, if required (see 3.4.3).

t. When only one drive system for ram or boom lift travel rate is acceptable, the particular drive required shall be as specified (see 3.4.3.2).

u. DC power operated infinitely variable lift rate, if different (see 3.4.3.2).

v. When only one drive system for reach on the ram or boom is acceptable, the particular drive required shall be as specified (see 3.4.3.3).

w. DC power operated infinite reach rates of travel, specify if (see 3.4.3.3).

x. Welding path accuracy, if different (see 3.4.4).

y. Voltage requirements, if different (see 3.4.5.).

z. Initial voltage, as specified (see 3.4.5.).

aa. Motors, if different (see 3.4.5.1).

bb. Control system, if different (see 3.4.5.2).

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cc. Worklight, if required (see 3.4.6).

dd. Size and capacity, if different (see 3.5).

ee. Specify physical size limitation, if required (see 3.5.1).

ff. Additional performance features, if required (see 3.6.1).

gg. Specify and fully describe optional equipment, if required (see 3.8).

hh. Specify applicable repair parts, if required (see 3.9).

ii. Lubrication plate, if required (see 3.10.1).

jj. Lubrication plate information, if different (see 3.10.1).

kk. Nameplate, if different (see 3.10.2).

ll. Inspection conditions, if different (see 4.3).

mm. First article inspection, if different (see 4.4).

nn. Quality conformance inspection, if different (4.5).

oo. Acceptance test, if different (see 4.6).

pp Additional performance tests, if required (see 4.8.3.1).

qq. Noise level acceptance test, if different (see 4.8.3.6).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-M-80173B	MIL-M-80173C

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

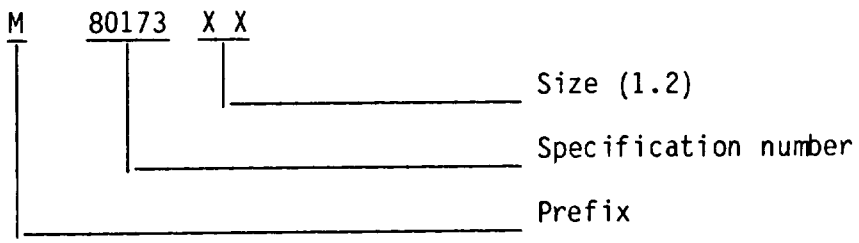
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

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6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation, the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PINs to be used for manipulators acquired to this specification are created as follows:



6.14 Subject Key word listing.

Circumferential seam welding
External seam welding
Horizontal seam welding
Longitudinal seam welding
Metal inert gas (MIG) welding
Tungsten inert gas (TIG) welding

Custodians:
Army - AL
Navy - SH
Air Force-99

Review activities:
Navy - AS, OS
Air Force - 84
DLA - GS

User activities:
Navy - MC

Preparing activity:

DLA - IP

Project (3436-0048)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: MIL-M-80173C	2. DOCUMENT DATE (YYMMDD): 92 02 10

3. DOCUMENT TITLE: MANIPULATORS, AUTOMATIC WELDING HEAD, STATIONARY BASE & TRAVELING RAM OR BOOM, EMD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG /Mr. Purdy	b. TELEPHONE (Include Area Code) (1) Commercial: (901)-775-6699 (2) AUTOVON: 683-6699
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114- 5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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